



CSE492 ENGINEERING PROJECT PROPOSAL FORM

Date: 29.01.2026

CSE 492 Registration Semester:

Make sure that you have registered to the CSE492 course via OBS before filling out this form.

Student Id: 20220702108	Tel.: 0546 293 72 84 E-Mail: resul.eryurt@std.yeditepe.edu.tr
Student Name, Surname, Signature:	Resul ERYURT
CSE 492 registration type (Mark one of the following choices): ☒ Regular ☐ Single Course (Tek Ders) ☐ Supplementary Course (Ek Ders) Extended	
This is a group project other members are listed: <u>Arda Egemen İPEK</u> <u>Ata Yağız GÜNER</u>	
Project Title	YuGoDa: Food Waste Reduction Application
Abstract (200 words)	This engineering project focuses on the design and implementation of the comprehensive full-stack architecture for YuGoDa, a platform dedicated to reducing food waste through a sustainable marketplace. The primary objective is to develop a robust backend system using Spring Boot and PostgreSQL, designed to handle secure user authentication, real-time inventory management for 'Surprise Bags,' and transactional logic. Simultaneously, the project involves constructing a responsive, cross-platform frontend using React.js and TypeScript, providing intuitive dashboards for both business vendors managing surplus stock and consumers seeking affordable meals. A critical component of this work involves the seamless integration of AI services into the application layer. While the core LLM training and prompt engineering are handled in parallel, this project focuses on embedding these intelligent features into the User Experience (UX). This includes developing the necessary API endpoints and UI components to facilitate the natural language chatbot and displaying hyper-personalized meal recommendations to the user. The final deliverable will be a scalable, high-performance web application that leverages modern software engineering principles to make the AI-driven circular economy accessible and functional for end-users.
Hardware constraints:(PC, Apple, mobile phone, other device, etc.) - PC/Mac/Linux computers with modern web browsers for accessing the web application. - Development machines: Personal laptops for the 3 people development team.	
Software constraints: (Operating System, Progr. Languages, Libraries, API's, IDE's etc) OS: Cross platform (Windows, macOS, Linux) Languages: JavaScript/TypeScript (React.js frontend), Java 17 (Spring Boot backend). Libraries: React Router, Axios, Redux Toolkit, Spring, Spring Data JPA, LangChain4j. Database: PostgreSQL. APIs: Google Maps API, OpenAI/Claude API (AI chatbot), Firebase Auth (Social Login and Authentication) Payment API. IDE: VS Code. DevOps: Docker, GitHub. Cloud: AWS.	
Evaluation criteria (Quality Measurements, Scalability, Error Rate, etc) Quality: Minimum %70unit test coverage, API response time under 500ms. Scalability: Support for 100-200 concurrent users, 3+ restaurant listings. Error Rate: Payment failure rate under 15%, Order processing errors under 10%. Performance: Page load time under 7 seconds, search results under 6 seconds. AI Accuracy: Achieving an F1-Score of 0.75 or higher in recommendation classification; Precision and Recall metrics will be used to evaluate the accuracy of the AI-powered assistant.	

<u>Design Methodology Diagrams</u>	
☒ Use Case Diagrams ☒ Class Diagrams ☒ Sequence Diagrams ☐ State Diagrams	
<u>Impact area</u>	
☒ Health ☒ Environmental ☒ Security ☒ Social /	
☒ Ethical Sustainability ☒ Productivity Other: _____	
Project Advisor Name, Surname, Signature:	Academic Advisor Name, Surname, Signature:
Gürhan KÜÇÜK	Esin ONBAŞIOĞLU